

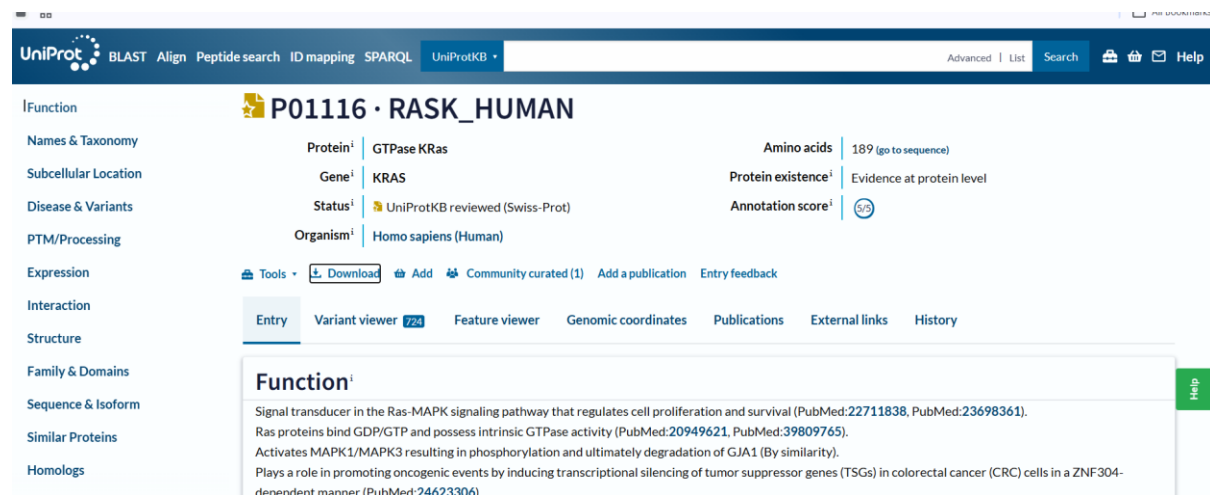
Task 1

Aim : To identify homologous proteins of human KRAS using BLASTP, evaluate sequence conservation across related species, and analyze percentage identity, query coverage, E-value, and alignment scores for key functional domains.

Introduction : The human *KRAS* gene encodes a small GTPase belonging to the Ras protein family, functioning as a molecular switch in the Ras-MAPK signaling cascade to regulate cell proliferation, differentiation, and survival. The canonical human KRAS protein comprises 189 amino acids and features a highly conserved N-terminal GTPase catalytic domain alongside a variable C-terminal hypervariable region (HVR) responsible for membrane targeting.

To evaluate the evolutionary conservation and sequence homology of human KRAS across organisms, the Basic Local Alignment Search Tool for proteins (BLASTP) was utilized. BLASTP aligns a query protein sequence against a target database using heuristic algorithms to detect statistically significant sequence similarities. Analyzing parameters such as Bit score, E-value, percent identity, and conservative substitutions allows for the identification of functionally indispensable motifs, including the phosphate-binding loop (P-loop) and switch regions critical for GTP hydrolysis.

Results:



The screenshot displays the UniProtKB entry for P01116, KRAS_HUMAN. The page is organized into a sidebar on the left with categories like Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence & Isoform, Similar Proteins, and Homologs. The main content area shows the protein's details, including its name (P01116 · RASK_HUMAN), protein name (GTPase KRas), gene name (KRAS), status (UniProtKB reviewed (Swiss-Prot)), and organism (Homo sapiens (Human)). It also provides amino acid count (189), protein existence evidence, and an annotation score. Below this, there are tabs for Entry, Variant viewer (224), Feature viewer, Genomic coordinates, Publications, External links, and History. The 'Function' section is expanded, showing the protein's role as a signal transducer in the Ras-MAPK signaling pathway, its ability to bind GDP/GTP and possess intrinsic GTPase activity, its role in activating MAPK1/MAPK3, and its involvement in promoting oncogenic events by inducing transcriptional silencing of tumor suppressor genes (TSGs) in colorectal cancer (CRC) cells.

Category	Value
Protein	GTPase KRas
Gene	KRAS
Status	UniProtKB reviewed (Swiss-Prot)
Organism	Homo sapiens (Human)
Amino acids	189 (go to sequence)
Protein existence	Evidence at protein level
Annotation score	4.5

Function
Signal transducer in the Ras-MAPK signaling pathway that regulates cell proliferation and survival (PubMed:22711838, PubMed:23698361). Ras proteins bind GDP/GTP and possess intrinsic GTPase activity (PubMed:20949621, PubMed:39809765). Activates MAPK1/MAPK3 resulting in phosphorylation and ultimately degradation of GJA1 (By similarity). Plays a role in promoting oncogenic events by inducing transcriptional silencing of tumor suppressor genes (TSGs) in colorectal cancer (CRC) cells in a ZNF304-dependent manner (PubMed:24623306).

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Standard Protein BLAST

BLASTP programs search protein databases using a protein query. [more...](#)

[Reset page](#)[Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Enter query sequence(s) in the text area. It automatically determines the format of the input. To allow this feature, certain conventions are required with regard to the input of identifiers. [more...](#)

Query subrange [?](#)

Enter coordinates for a subrange of the query sequence. The BLAST search will apply only to the residues in the range. Sequence coordinates are from 1 to the sequence length. The range includes the residue at the To coordinate. [more...](#)

From

To

Or, upload file

Choose File

KRAS_HUMAN.fasta [?](#)

Use the browse button to upload a file from your local disk. The file may contain a single sequence or a list of sequences. The data may be either a list of database accession numbers, NCBI gi numbers, or sequences in FASTA format.

Job Title

Enter a descriptive title for your BLAST search [?](#)

This title appears on all BLAST results and saved searches.

Organism

Optional

Select the sequence database to run searches against. No BLAST database contains all the sequences at NCBI. BLAST databases are organized by informational content (nr, RefSeq, etc.) or by sequencing technique (WGS, EST, etc.). [more...](#)

☐ exclude

[Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Start typing in the text box, then select your taxid. Use the "plus" button to add another organism or group, and the "exclude" checkbox to narrow the subset. The search will be restricted to the sequences in the database that correspond to your subset.

Exclude

Optional

☐ Models (XMP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Entrez® Query

Optional

Create custom database

Enter an Entrez query to limit search [?](#)

You can use Entrez query syntax to search a subset of the selected BLAST database. This can be helpful to limit searches to molecule types, sequence lengths or to exclude organisms. [more...](#)

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST

Search using Blastp (protein-protein BLAST)

☒ Show results in a new window

+ Algorithm parameters

Page

Reset

Taxonomy

Download

[Genome Data Viewer](#) - aligned genomic context

Hit	Protein Description	Accession	Max Score	Total Score	Query Cover	E-value	Percent Identity
1	GTPase KRas isoform X1 [<i>Canis lupus dingo</i>]	XP_048958751.1	390	390	100%	2×10^{-136}	98.94%
2	GTPase KRas isoform X4 [<i>Canis lupus dingo</i>]	XP_048958756.1	391	391	100%	2×10^{-136}	98.94%
3	unnamed protein product, partial [<i>Ovis aries</i>]	VCX04206.1	390	390	100%	6×10^{-136}	98.94%

Hit	Protein Description	Accession	Max Score	Total Score	Query Cover	E-value	Percent Identity
4	UBE2L3/KRAS fusion protein [<i>Homo sapiens</i>]	AEA35014.1	391	391	100%	1×10^{-135}	100.00%

Interpretation

The BLASTP analysis revealed exceptionally high sequence conservation for human KRAS across mammalian homologs. The top alignment (*Canis lupus dingo*, 98.94% identity, $E\text{-value} = 2 \times 10^{-136}$) demonstrates 100% identity across the primary catalytic GTPase domain (residues 1–120), preserving the critical P-loop motif (GAGGVGKS) responsible for nucleotide binding. Minor conservative amino acid substitutions (such as Ser175 and Ile187) were restricted strictly to the C-terminal hypervariable region, reflecting evolutionary tolerance for variation in non-catalytic membrane-anchoring zones while keeping the enzymatic core invariant.

Conclusion

BLASTP successfully identified close homologs of human KRAS. The extreme sequence conservation—evidenced by high Bit scores and near-zero E-values—underscores the strong evolutionary constraint on the GTPase domain necessary for maintaining vital intracellular MAPK signaling pathways.

Once you enter those top 5 hits into the table, Task 1 is 100% complete. You can now move on to Task 2 (Clustal Omega MSA) whenever you're ready!